DOI: 10.1002/adsc.201300602

Palladium(0)-Catalyzed Regioselective Synthesis of Macrocycles from Allenes with a Nucleophilic Functionality and Organic Iodides

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Received: July 6, 2013; Revised: August 22, 2013; Published online: November 11, 2013

Supporting information for this article is available on the WWW under http://dx.doi.org/10.1002/adsc.201300602.

Abstract: A highly regioselective synthesis of 15–20-membered macrocycles *via* Pd(0)-catalyzed coupling cyclization of allenes bearing a carbon or nitrogen nucleophilic functionality and organic iodides is reported. After oxidative addition and carbopalladation, *anti*-π-allyl palladium intermediates were formed, which was followed by the exclusive intramolecular nucleophilic attack at the terminal carbon atom of the allene moiety to afford products with yields ranging from 44–92% and *E/Z* ratios of up to 97/3.

Keywords: allenes; carbopalladation; cyclization; macrocycles

Macrocycles exist widely in natural products with remarkable biological activities.[1] The most commonly used approaches to construct macrocycles include ring-closing metathesis,[2] macrolactonization,[3] macrolactamization,[4] and multi-component macrocyclization.[5] In spite of all these developed methods, facile and efficient new methods are still strongly in need.

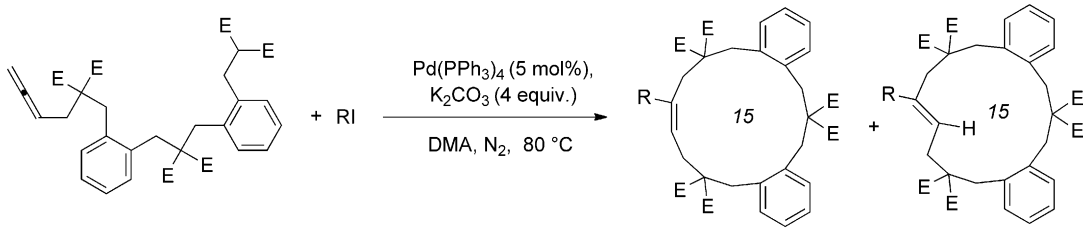
Nowadays, allenes have been playing a more and more important role in organic synthesis due to their interesting chemical properties associated with the two cumulated double bonds.[6] Transition metal-catalyzed cyclization reactions of functionalized allenes in the presence of organic halides have become very useful methodologies to construct cyclic compounds.[7] However, there are only rare reports concerning the synthesis of macrocycles *via* allenes.[8] It was noted that the high dilution technique was applied to increase the yields of macrocycles in both cases. In our previous work, we have reported highly regio- and stereoselective syntheses of 9- to 12-membered cyclic compounds from the Pd(0)-catalyzed cyclization of allenes having various carbon and nitrogen nucleophilic functionalities.[9] In this paper, we wish to report our recent observation on the construction of 15- to 20-membered macrocycles with high regioselectivity in normal concentrations.

Initially, we conducted the reaction of **1a** with PhI under the catalysis of $Pd(PPh_3)_4$ (5 mol%) with K_2CO_3 (4 equiv.) in DMA, 80 °C.[9] Fortunately, after 12 h, the reaction succeeded to afford the expected product (*E*)-**3aa** in 82% yield with a decent *E/Z* ratio (95/5) (entry 1, Table 1). The structure of (*E*)-**3aa** especially the C=C bond on the ring was further established by an X-ray diffraction study (Figure 1).[10] Then these standard conditions were applied to investigate the reaction of **1a** with other different organic iodides (Table 1). Not only electron-rich (entries 2 and 3, Table 1) but also electron-deficient substituents (entries 4–10, Table 1) could all be introduced into aryl iodides: the corresponding 15-membered rings were isolated in moderate to good yields (66–80%) with high *E/Z* ratios (94/6–97/3). Interestingly, when 4-iodonitrobenzene (**2k**) was used, the corresponding products (*E*)-**3ak** and (*Z*)-**3ak** were afforded in 79% and 7% yields, respectively (entry 11, Table 1). 2-Thienyl iodide (**2l**) could also react with **1a**, which afforded (*E*)-**3al** in 62% yield and (*Z*)-**3al** in 7% yield (entry 13, Table 1). In addition, 1-(*E*)-hexenyl iodide (**2m**) may also be applied, resulting in an isolated yield of 35% for (*E*)-**3am** and 25% for (*Z*)-**3am**, respectively (entry 14, Table 1). We also investigated the reaction of **1k** on a 1-g scale under the optimized conditions and obtained an 80% isolated yield of (*E*)-**3ak** and a 4% yield of (*Z*)-**3ak** (entry 12, Table 1).

To our surprise, when we tried the reaction of **1a** with 4-iodoanisole (**2n**), in addition to the expected macrocycles (*E*)-**3an** in 57% yield and (*Z*)-**3an** in 3% yield (by ^{1}H NMR), a 9% yield of the phenyl-substituted product (*E*)-**3aa** was also isolated [Eq. (1)]. The possible mechanism of the formation of (*E*)-**3aa** is

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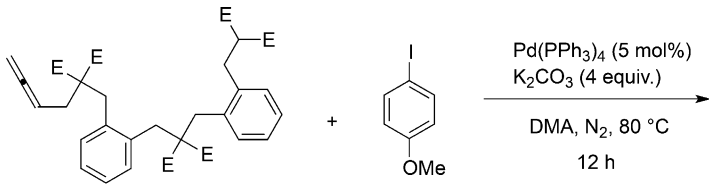
Table 1. Pd(0)-catalyzed coupling–cyclization of **1a** with different organic iodides under the standard conditions.[a]



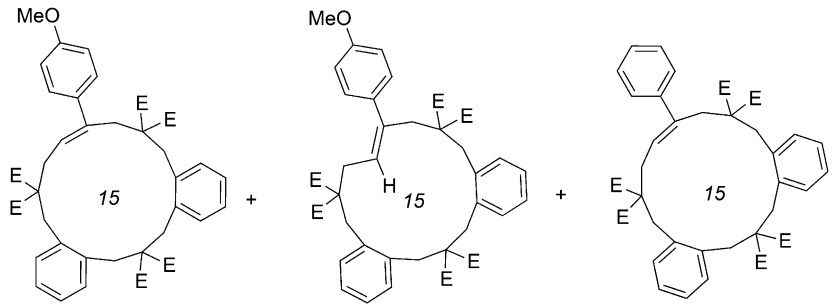
1a (E = CO_2Me) **2** (2.0 equiv.) (*E*)-**3** (*Z*)-**3**

Entry	RI (**2**) R =	Time [h]	(*E*/*Z*)[b]	Yield of **3**[c] [%] (*E*)-**3**	(*Z*)-**3**
1	Ph (**2a**)	12	95/5	82 (**3aa**)	–[d]
2	*p*-MeC_6H_4 (**2b**)	11.5	94/6	66 (**3ab**)	–[d]
3	*m*-MeC_6H_4 (**2c**)	12	95/5	72 (**3ac**)	–[d]
4	*p*-IC_6H_4 (**2d**)	12	97/3	71 (**3ad**)	–[d]
5	*p*-BrC_6H_4 (**2e**)	11	96/4	80 (**3ae**)	–[d]
6	*p*-ClC_6H_4 (**2f**)	12	96/4	74 (**3af**)	–[d]
7	*p*-$MeOOCC_6H_4$ (**2g**)	13	95/5	79 (**3ag**)	–[d]
8	*p*-$BnOOCC_6H_4$ (**2h**)	12	95/5	78 (**3ah**)	–[d]
9	*p*-$MeOCC_6H_4$ (**2i**)	14	96/4	74 (**3ai**)	–[d]
10	*p*-NCC_6H_4 (**2j**)	16	96/4	74 (**3aj**)	–[d]
11	*p*-$NO_2C_6H_4$ (**2k**)	12	90/10	79 (**3ak**)	7[e] (**3ak**)
12[f]	*p*-$NO_2C_6H_4$ (**2k**)	12	95/5	80 (**3ak**)	4[g] (**3ak**)
13	2-thienyl (**2l**)	16	86/14	62 (**3al**)	7 (**3al**)
14[h]	(*E*)-1-hexenyl (**2m**)	72	59/41	35 (**3am**)	25 (**3am**)

[a] The reaction was conducted with 0.067 M of **1a**.
[b] Determined by ^1H NMR analysis of the crude products with CH_2Br_2 as the internal standard.
[c] Isolated yield.
[d] The amount of pure *Z*-isomer was too small for isolation.
[e] This isolated *Z*-isomer was contaminated with 11% *E*-isomer.
[f] The reaction was conducted on a 1-g scale.
[g] This isolated *Z*-isomer was contaminated with 8% *E*-isomer.
[h] **2m** (4 equiv.) was used.



1a **2n**



(*E*)-**3an**, 57% (*Z*)-**3an**, 3% by ^1H NMR (*E*)-**3aa**, 9% (1)

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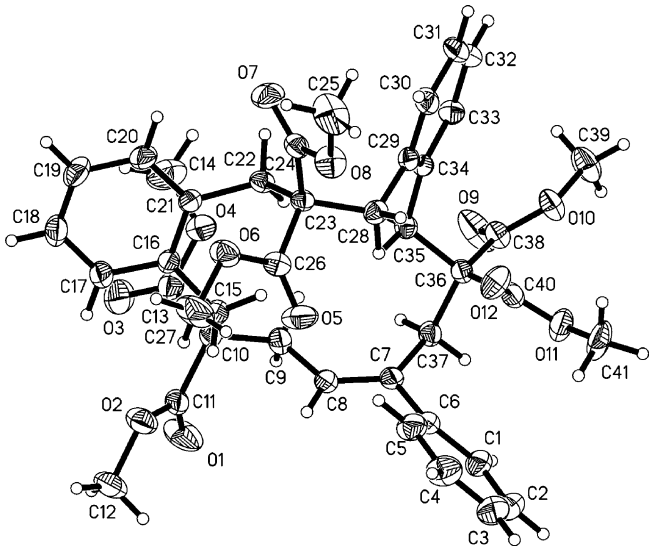


Figure 1. ORTEP representation of (*E*)-**3aa**.

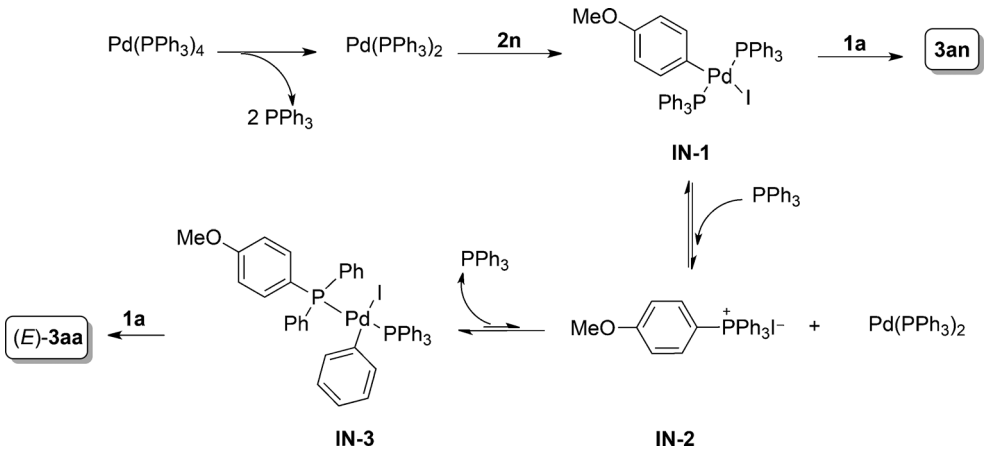
shown in Scheme 1. The oxidative addition of $Pd(PPh_3)_2$ with **2n** generates the 4-methoxyphenylpalladium iodide intermediate **IN-1**. Then the reductive elimination between PPh_3 and the 4-methoxyphenyl group in intermediate **IN-1** generates phosphonium iodide **IN-2**, which undergoes the oxidative addition with $Pd(PPh_3)_2$ to generate a new intermediate **IN-3**, causing the cleavage of the P–Ph bond.[11] Subsequent reaction with **1a** afforded the unexpected phenyl-substituted (*E*)-**3aa**.

To our delight, when a nitrogen nucleophile was introduced, i.e., **1b**, the expected azamacrocycle (*Z*)-**3ba** may also be obtained in 80% yield with a *Z*/*E* ratio of 96/4. Interestingly, the reaction of **1c** gave the corresponding nitrogen-tethered heteromacrocycle (*E*)-**3ca** in 44% yield as the only product. Similarly, an oxygen tether may also be applied: the reaction of **1d** afforded 39% isolated yield of (*E*)-**3da** and 8% isolated yield of (*Z*)-**3da** (Scheme 2).

Subsequently, we conducted the reaction of **1e** containing only one C_6H_4 unit with PhI under the previous standard conditions. Unfortunately, after 24 h, the reaction afforded only 20% yield of macrocycle **3ea** with 38% recovery of **1e** probably due to the increased flexibility of the carbon chain in **1e** (entry 1, Table 2). Replacing $Pd(PPh_3)_4$ (5 mol%) with $Pd_2(dba)_3{\cdot}CHCl_3$ (2.5 mol%) and TFP [10 mol%, TFP = tris(2-furyl)phosphine], the reaction afforded 25% yield of macrocycle **3ea** albeit with 28% recovery of **1e** after 24 h (entry 2, Table 2). Interestingly, with this catalytic system, after screening a series of bases, we found that the reaction using K_3PO_4 afforded an 85% yield of macrocycle **3ea** with an *E*/*Z* selectivity of 85/15 (entries 3–5, Table 2). The reaction in THF or CH_3CN afforded **3ea** in much lower yields (entries 6 and 7, Table 2).

Thus, the following conditions [$Pd_2(dba)_3{\cdot}CHCl_3$ (2.5 mol%), TFP (10 mol%), and K_3PO_4 (4.0 equiv.) in DMA, 80 °C] were established as the optimized conditions for further study with the typical results summarized in Table 3. The following issues should be noted: (i) yields of macrocycles **3** range from 51% to 92%; (ii) all the corresponding products were inseparable *E*/*Z* mixtures except **3aa**, and the *E*/*Z* ratios range from 60/40 to 95/5; (iii) decreasing the rigidity or increasing the size of the carbon chain in the substrates may cause a decrease of the *E*/*Z* ratio.

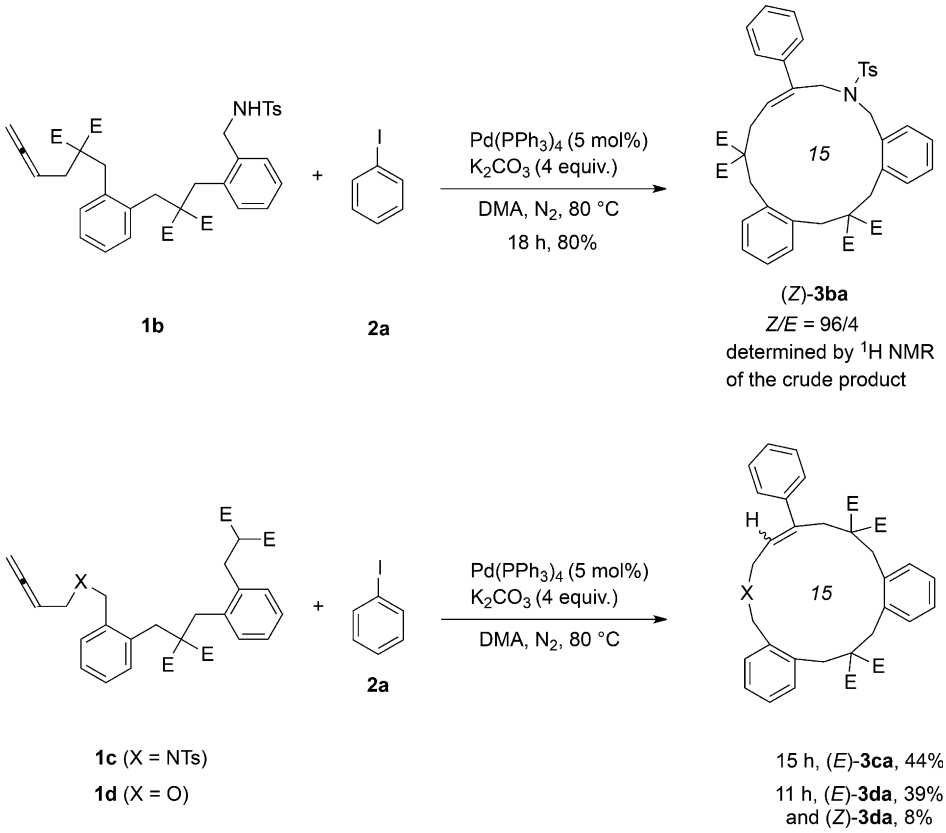
Then we conducted the hydrogenation of products **3** under the conditions of Pd/C (10 mol%) and H_2 (30 bar) at room temperature. Interestingly, we found that the solvent played an important role. For **3aa**, **3ea**, and **3ga**, EtOAc/MeOH (1/1) was the proper mixed solvent (entries 1, 2, 4, Table 3), but it did not work for **3fa**. Pleasantly, **3fa** could be completely hydrogenated when CH_2Cl_2/MeOH (1/1) was used instead of EtOAc/MeOH (1/1) (entry 3, Table 3).



Scheme 1. Mechanism for the formation of (*E*)-**3aa** in the reaction of **1a** with **2n**.

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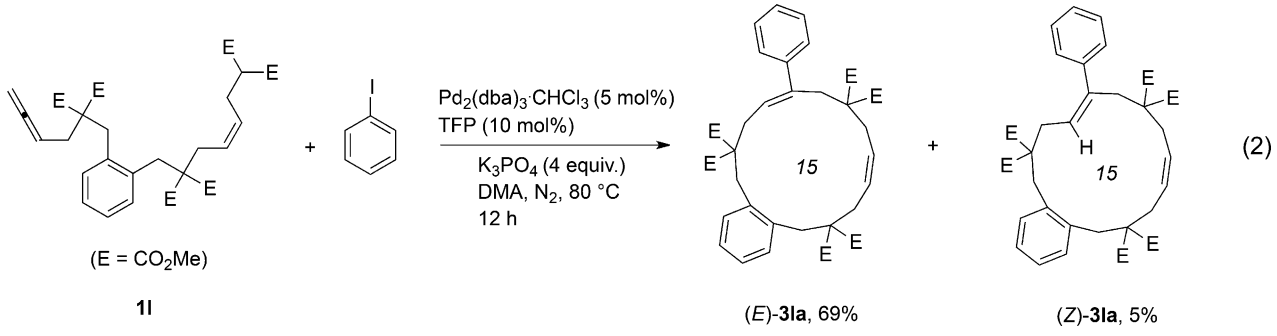
Scheme 2. Synthesis of aza- and oxa-macrocycles.

Unfortunately, neither of the mentioned mixed solvents was effective for the hydrogenation of **3ha** (entries 1 and 2, Table 4). Then we tried to add CH_2Cl_2 (0.1 mL) to EtOAc/MeOH (2.0 mL/2.0 mL) with ~95% of **3ha** being recovered (entry 3, Table 4); Luckily, when only 0.5 equiv. of CH_2Cl_2 was added, 5% of **3ha** was recovered after 58 h (entry 4, Table 4); Replacing CH_2Cl_2 (0.5 equiv.) with $CHCl_3$ (0.5 equiv.), the complete hydrogenation of **3ha** succeeded (entry 5, Table 4). Increasing the amount of $CHCl_3$ to 1.0 equiv. could shorten the reaction time to 48 h (entry 6, Table 4).

Subsequently, we found that this mixed solvent [$CHCl_3$ (1.0 equiv) and EtOAc/MeOH (1/1)] was also suitable for **3ia** and **3ja** (entries 6 and 7, Table 3) and the amount of $CHCl_3$ should be increased to 2.0 equiv. for **3ka** (entry 8, Table 3).

In addition, the substrate **1l** with a *Z* C=C bond could also react with PhI to afford a 69% isolated yield of (*E*)-**3la** and a 5% isolated yield of (*Z*)-**3la** by chromatography on silica gel with a purity of only ~86% [Eq. (2)].

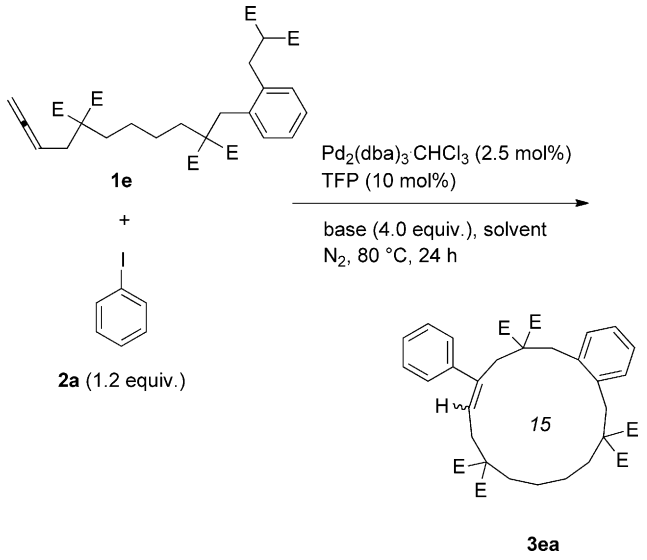
In conclusion, we have developed an efficient method for the synthesis of 15- to 20-membered macrocycles *via* Pd(0)-catalyzed coupling–cyclization reaction of functionalized allenes with organic iodides in high regioselectivity without applying a high dilution technique. The rigidity and size of the carbon chain in the substrates have a great influence on the stereoselectivity of the formed macrocycles. The re-



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Table 2. Effects of solvent and base in Pd(0)-catalyzed coupling–cyclization of **1e** with PhI.[a]



Entry	Base	Solvent	Yield [%][b] of **3ea** (*E*/*Z*)	Recovery of **1e** [%][b]
1[c]	K_2CO_3	DMA	20 (45/55)	38
2	K_2CO_3	DMA	25 (66/34)	28
3	Na_2CO_3	DMA	0	72
4	Cs_2CO_3	DMA	34 (82/18)	0
5[d]	K_3PO_4	DMA	85 (85/15)	0
6	K_3PO_4	THF	55 (92/8)	0
7	K_3PO_4	CH_3CN	19 (72/28)	7

[a] The reaction was conducted with 0.067 M of **1e** in 0.1–0.2 mmol scale.
[b] Determined by ^{1}H NMR analysis of the crude products with CH_2Br_2 as the internal standard.
[c] $Pd(PPh_3)_4$ (5 mol%) was used instead of $Pd_2(dba)_3{\cdot}CHCl_3$ (2.5 mol%) and TFP (10 mol%).
[d] The reaction time was 12 h.

gioselectivity may be rationalized by the presence of an R group in the to-be-formed allylic Pd moiety,[12] which is in its *anti*-form to avoid the steric repulsion between the R group and the carbon chain bearing the nucleophilic moiety (Scheme 3).[13] Further studies in this area are being pursued in our laboratory.

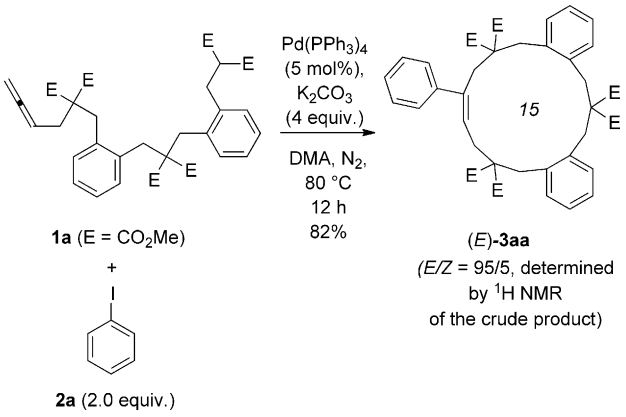
Experimental Section

General Experimental Methods

^{1}H and ^{13}C nuclear magnetic resonance spectra were recorded on an instrument operated at 300 MHz for ^{1}H and 75 MHz for ^{13}C. Deuterochloroform ($CDCl_3$) was used as solvent in all NMR experiments. Chemical shifts (δ) are given in parts per million (ppm). Infrared spectra were recorded from films of pure samples on sodium chloride plates for liquids or in the form of KBr discs for the solid samples. Mass and HR-mass spectra were carried out in the ESI mode. Thin layer chromatography was performed on pre-coated glassback plates and visualized with UV light at 254 nm. Flash column chromatography was performed on silica gel. THF was refluxed in the presence of sodium wire using diphenyl ketone as indicator and distilled right before use. DMA used was stirred with CaH_2 for 12 h at 80 °C and distilled under vacuum before use. K_2CO_3 was bought from Sinopharm Chemical Reagent Co., Ltd. $Pd_2(dba)_3{\cdot}CHCl_3$, Pd/C and K_3PO_4 were bought from Alfa Aesar. TFP was bought from Aldrich.

General Procedure for Pd(0)-Catalyzed Coupling–Cyclization of 1a with Different Organic Iodides 2 under Standard Conditions

Preparation of (*E*)-3aa; Typical Procedure I

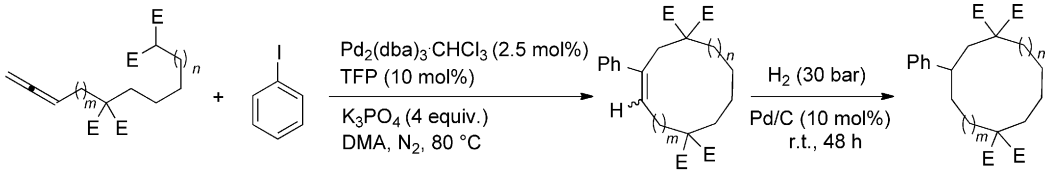


To a flame-dried Schlenk tube containing K_2CO_3 (109.8 mg, 0.8 mmol) were added $Pd(PPh_3)_4$ (12.0 mg, 0.01 mmol), **2a** (81.7 mg, 0.4 mmol), DMA (0.5 mL), a solution of **1a** (*w*/*w* = 9.54% in DMA) (1.3637 g, 0.2 mmol), and DMA (1.2 mL) sequentially under a nitrogen atmosphere. The reaction was complete after being stirred at 80 °C for 12 h as monitored by TLC (eluent: petroleum ether/ethyl acetate/dichloromethane = 5/1/1). The resulting mixture was quenched with 10 mL of H_2O, extracted with ether (30 mL × 3), washed with water and brine, and dried over anhydrous Na_2SO_4. After filtration and evaporation of the solvent, the *E*/*Z* ratio (95/5) was determined by ^{1}H NMR analysis of the crude product with CH_2Br_2 as the internal standard. Chromatography on silica gel [eluent: dichloromethane/ethyl acetate (40/1)] afforded (*E*)-**3aa** as a white solid; yield: 119.7 mg (82%); mp 215–216 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$); δ = 7.46–6.94 (m, 13 H, ArH), 5.61 (t, *J* = 6.6 Hz, 1 H, CH=), 3.94–3.31 (22 H, 4 × Me and 5 × CH_2), 3.16 (d, *J* = 6.6 Hz, 2 H, CH_2), 3.10 (s, 6 H, 2 × Me); ^{13}C NMR (75 Hz, $CDCl_3$): δ = 172.0, 171.3, 170.9, 142.6, 138.9, 136.1, 136.0, 135.8, 135.5, 130.7, 130.1, 130.0, 129.5, 129.3, 127.8, 127.6, 127.0, 126.8, 126.4, 126.3, 126.2, 60.5, 59.5, 58.8, 52.34, 52.30, 51.7, 36.7, 36.5, 36.1, 36.0, 35.8, 35.7; IR (KBr): ν = 2952, 1728, 1492, 1434, 1337, 1244, 1205, 1116, 1057 cm^{-1}; MS (ESI): *m*/*z* = 813 ($M^+ + K + C_2H_5OH$), 767 ($M_+ + K$), 751 ($M_+ + Na$), 729 ($M_+ + H$); anal. calcd. for $C_{41}H_{44}O_{12}$ (%): C 67.57, H 6.09; found: C 67.26, H 6.04.

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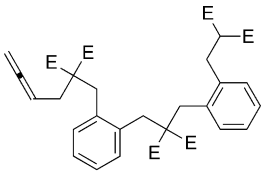
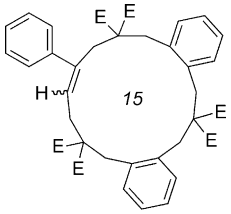
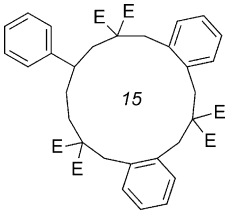
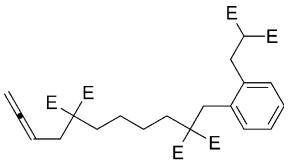
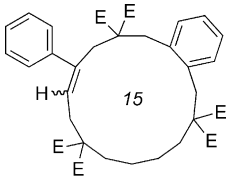
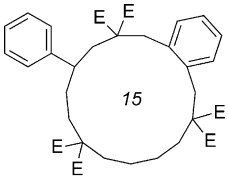
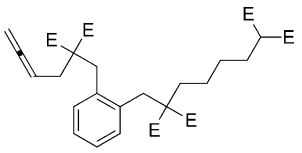
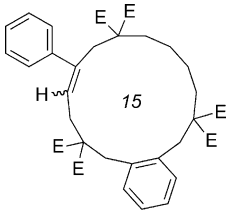
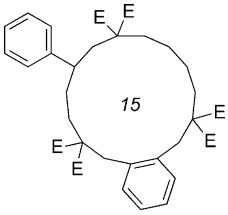
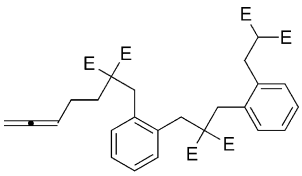
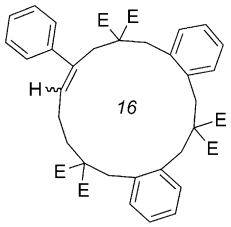
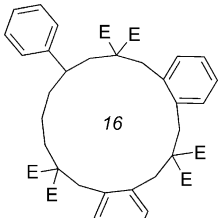
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Table 3. Pd(0)-catalyzed coupling–cyclization of different substrates **1** with PhI and the hydrogenation of products **3**.[a]



1 **2a** **3** **4**

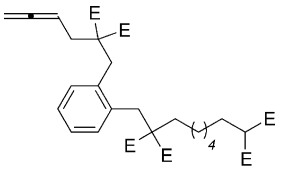
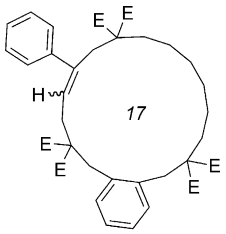
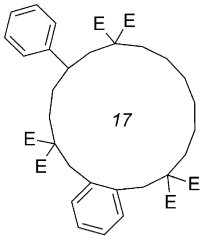
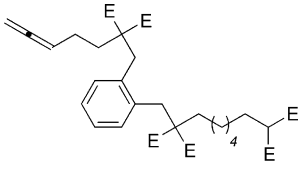
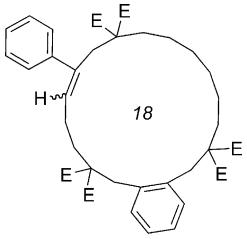
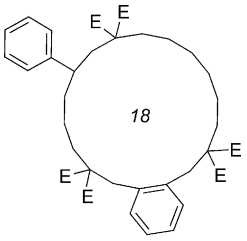
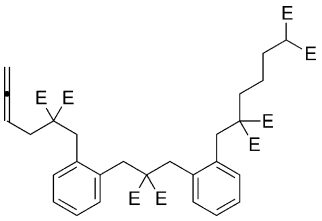
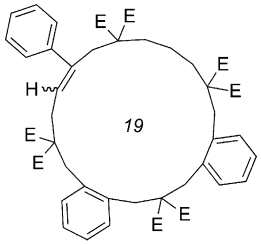
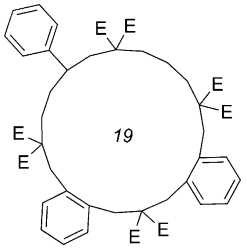
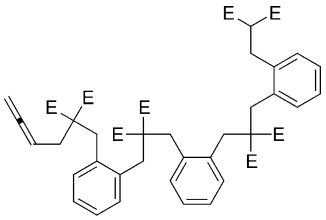
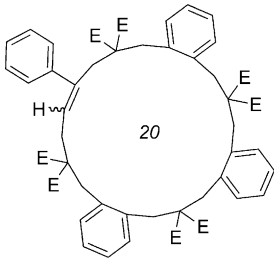
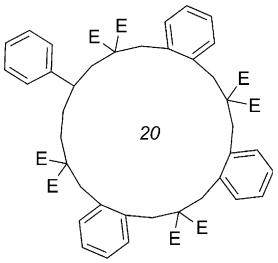
E = CO_2Me
m = 1, 2, *n* = 6–11

Entry	Substrate **1**	Product **3** (time, isolated yield)	Product **4** (solvents, isolated yield from **3**)
1	 **1a**[b]	 (*E*/*Z*)-**3aa** = 95/5 12 h, 92%	 **4aa** EtOAc/MeOH = 1/1 99%
2	 **1e**	 (*E*/*Z*)-**3ea** = 85/15 12 h, 77%	 **4ea** EtOAc/MeOH = 1/1 100%
3	 **1f**	 (*E*/*Z*)-**3fa** = 81/19 14 h, 65%	 **4fa** CH_2Cl_2/MeOH = 1/1 100%
4	 **1g**[b]	 (*E*/*Z*)-**3ga** = 77/23 12 h, 80%	 **4ga** EtOAc/MeOH = 1/1 99%

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Table 3. (Continued)

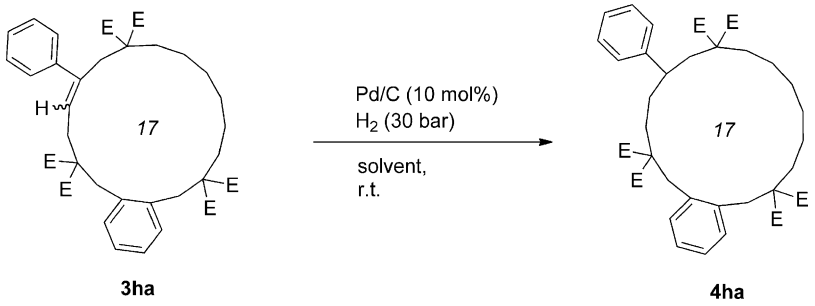
Entry	Substrate **1**	Product **3** (time, isolated yield)	Product **4** (solvents, isolated yield from **3**)
5	 **1h**	 (*E*/*Z*)-**3ha** = 63/37 12 h, 61%	 **4ha** $CHCl_3$ (1.0 equiv.) EtOAc/MeOH = 1/1 99%
6	 **1i**	 (*E*/*Z*)-**3ia** = 60/40 12 h, 51%	 **4ia** $CHCl_3$ (1.0 equiv.) EtOAc/MeOH = 1/1 99%
7	 **1j**	 (*E*/*Z*)-**3ja** = 72/28 12 h, 56%	 **4ja** $CHCl_3$ (1.0 equiv.) EtOAc/MeOH = 1/1 100%
8	 **1k**	 (*E*/*Z*)-**3ka**=68/32 12 h, 52%	 **4ka** $CHCl_3$ (2.0 equiv.) EtOAc/MeOH = 1/1 99%

[a] The reaction was conducted with 0.067 M of **1**.
[b] $Pd(PPh_3)_4$ (5 mol%) was used instead of $Pd_2(dba)_3{\cdot}CHCl_3$ (2.5 mol%) and TFP (10 mol%).

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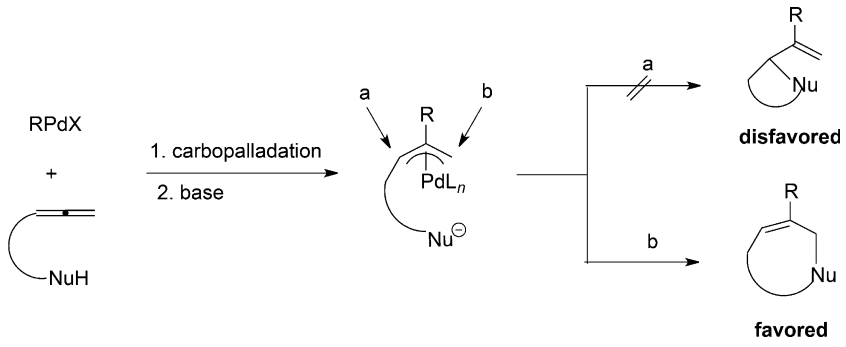
Table 4. Selections of mixed solvents for the hydrogenation of **3ha**.



3ha **4ha**

Entry	Scale [mmol]	Solvent	Time [h]	Recovery of **3ha**[a]
1	0.115	EtOAc (2.0 mL)/MeOH (2.0 mL)	44	100%
2	0.175	CH_2Cl_2 (2.5 mL)/MeOH (2.5 mL)	44	100%
3	0.109	CH_2Cl_2 (0.1 mL)/EtOAc (2.0 mL)/MeOH (2.0 mL)	37	95%
4	0.121	CH_2Cl_2 (0.5 equiv.)/EtOAc (2.0 mL)/MeOH (2.0 mL)	58	5%
5	0.122	$CHCl_3$ (0.5 equiv.)/EtOAc (2.0 mL)/MeOH (2.0 mL)	58	0
6	0.117	$CHCl_3$ (1.0 equiv.)/EtOAc (2.0 mL)/MeOH (2.0 mL)	48	0

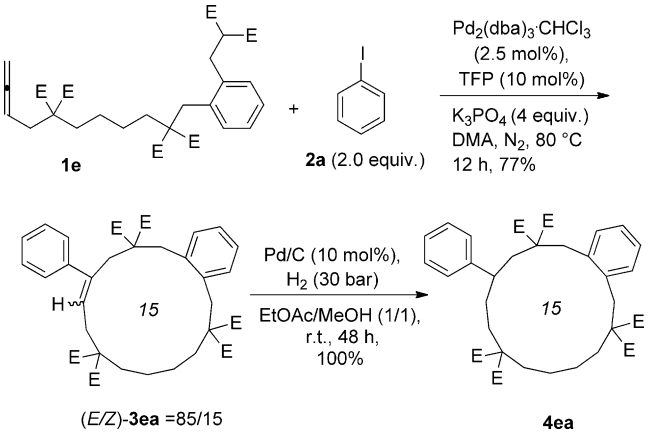
[a] Determined by ^{1}H NMR analysis of the crude products with CH_2Br_2 as the internal standard.



Scheme 3. Explanation of the regioselectivity.

General Procedure for Pd(0)-Catalyzed Coupling–Cyclization of Different Substrates 1 with PhI and the Hydrogenation of Unseparated Products 3

Preparation of 4ea; Typical Procedure II



To a flame-dried Schlenk tube containing K_3PO_4 (1315.2 mg, 6.2 mmol) were added $Pd_2(dba)_3{\cdot}CHCl_3$ (40.4 mg, 0.039 mmol), TFP (36.1 mg, 0.155 mmol), **2a** (631.9 mg, 3.10 mmol)/DMA (3.3 mL), and **1e** (938.4 mg, 1.55 mmol)/DMA (20.0 mL) sequentially under a nitrogen atmosphere. The reaction was complete after being stirred at 80 °C for 12 h as monitored by TLC (eluent: petroleum ether/ethyl acetate/dichloromethane = 5/1/1). The resulting mixture was quenched with 20 mL of H_2O, neutralized with dilute hydrochloric acid to observe a clear solution, extracted with ethyl acetate (50 mL × 3), washed with water and brine, and dried over anhydrous Na_2SO_4. After filtration and evaporation of the solvent, chromatography on silica gel (eluent: dichloromethane/ethyl acetate = 40/1) afforded (*E/Z*)-**3ea**, which was submitted to the next step without further characterization; yield: 817.0 mg (77%, *E/Z* = 85/15).

To an autoclave was added Pd/C (dry, 10%, 14.9 mg, 0.014 mmol), **3ea** (94.7 mg, 0.139 mmol), ethyl acetate (2.0 mL) and methanol (2.0 mL) sequentially. Then the mixture was stirred under an H_2 (30 bar) atmosphere at room temperature for 48 h, and the resulting mixture was filtered through a short pad of silica gel eluted with 30 mL of ethyl

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acetate. The filtrate was concentrated under vacuum then followed by vacuum drying at 80 °C to afford pure **4ea** as a white solid; yield: 95.1 mg (100%); mp 195–196 °C (*n*-hexane/DCM); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.34–7.23 (m, 2H), 7.23–7.04 (m, 6H), 7.01–6.93 (m, 1H), 3.80–3.40 (m, 18H), 3.28 (d, J = 14.7 Hz, 1H), 3.02 (s, 3H, Me), 2.68–2.51 (m, 2H), 2.50–2.38 (m, 1H), 2.02–1.71 (m, 6H), 1.68–1.08 (m, 6H); ^{13}C NMR (75 Hz, $CDCl_3$): δ = 172.0, 171.7, 171.6, 170.6, 144.2, 135.8, 135.6, 130.0, 129.5, 128.3, 127.6, 126.44, 126.36, 59.2, 58.1, 57.0, 52.3, 52.23, 52.16, 52.0, 51.5, 42.1, 41.8, 35.2, 34.7, 32.1, 31.9, 29.8, 28.5, 24.9, 23.3; IR (KBr): ν = 2951, 1732, 1498, 1437, 1253 cm^{-1}; MS (ESI): m/z = 705 (M^+ + Na), 700 (M^+ + NH_4), 683 (M^+ + H); anal. calcd. for $C_{37}H_{46}O^{12}$ (%): C 65.09, H 6.79; found: C 64.78, H 6.80.

Acknowledgements

*Financial support from the National Natural Science Foundation of China (21232006) and the National Natural Basic Research Program of China (2011CB808700) is greatly appreciated. We thank Mr. Ruizhi Lv in this group for reproducing the preparation of (E)-**3ae**, (E)-**3am**, (Z)-**3am** in* Table 1 *and **4ha** in* Table 3. *Shengming Ma is a Qiu Shi Adjunct Professor at Zhejiang University.*

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[10] Crystal data for compound (*E*)-**3aa**: $C_{41}H_{44}O_{12}$, MW = 728.76, monoclinic, space group *P*21/*c*, final *R* indices [I > 2σ(I)], *R1* = 0.0546, *wR2* = 0.1162, *R* indices (all data) *R1* = 0.1064, *wR2* = 0.1406, *a* = 12.4112(7) Å, *b* = 15.2494(8) Å, *c* = 20.5091(13) Å, α = 90.00°, β = 104.346(6)°, γ = 90.00°, *V* = 3760.6(4) Å^3, *T* = 293(2) K, *Z* = 4, reflections collected/unique 23098/8174 (R_{int} = 0.0384), number of observations [> 2σ(I)] 4810, parameters: 495. CCDC 921122 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre *via* www.ccdc.cam.ac.uk/data_request/cif.

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